

A Randomized Clinical Trial for Meal Bolus Decision Using Learning-based Control in Adults With Type 2 Diabetes

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Abstract

Context: We propose an artificial-pancreas-like algorithm (AP-A) that could automatically determine the preprandial insulin dose based on intermittently scanned continuous glucose monitoring (isCGM) data trajectories in multiple dose injection (MDI) therapy.

Objective: We aim to determine whether preprandial insulin dose adjustments guided by the AP-A are as effective and safe as physician decisions.

Methods: We performed a randomized, single-blind, clinical trial at a tertiary, referral hospital in Beijing, China. Type 2 diabetes participants were eligible if they were aged 18 years or older, with a glycated hemoglobin A_{1c} of 8.0% or higher. Eligible participants were randomly assigned (1:1) to the AP-A arm supervised by physician and the conventional physician treatment arm. The primary objective was to compare percentage time spent with sensor glucose level in 3.9 to 10.0 mmol/L (TIR) between the 2 study arms. Safety was assessed by the percentage time spent with sensor glucose level below 3.0 mmol/L (TBR).

Results: A total of 140 participants were screened, of whom 119 were randomly assigned to the AP-A arm (n = 59) or physician arm (n = 60). The TIR achieved by the AP-A arm was statistically noninferior compared with the control arm (72.4% [63.3%–82.1%] vs 71.2% [54.9%–81.4%]), with a median difference of 1.33% (95% CI, –6.00 to 10.94, noninferiority margin –7.5%). TBR was also statistically noninferior between the AP-A and control arms (0.0% [0.0%–0.0%] vs 0.0% [0.0%–0.0%]), respectively; median difference (95% CI, 0.00% [0.00%–0.00%], noninferiority margin 2.0%).

Conclusion: The AP-A–supported physician titration of preprandial insulin dosage offers noninferior glycemic control compared with optimal physician care in type 2 diabetes.

Key Words: insulin treatment, type 2 diabetes, artificial pancreas, multiple daily injections, glucose control

Abbreviations: AP-A, artificial-pancreas-like algorithm; HbA_{1c}, glycated hemoglobin A_{1c}; HDL, high-density lipoprotein; isCGM, intermittently scanned continuous glucose monitoring; LDL, low-density lipoprotein; MDI, multiple daily injections; SBP, systolic blood pressure; TBR, time below range; TIR, time in range.

Prevalence of diabetes continues to increase worldwide and there is an ever-increasing number of patients who require insulin therapy (1). Proper adjustment of insulin doses is crucial to euglycemic control, but remains challenging both to patients and care providers (2). The availability of new technologies such as sensor augment insulin pump and hybrid closed-loop systems has further advanced insulin management without increase in hypoglycemia while achieving euglycemic control (3). However, less than 1% of diabetes patients receiving insulin therapy use insulin pumps and/or hybrid closed-loop systems. The escalating costs and challenges associated with implementing these technologies could pose substantial obstacles in real-world clinical practice (1). In this regard, it is crucial to promptly integrate the intelligent

closed-loop control algorithms employed in artificial pancreas (AP) systems into the more cost-effective and widely used multiple daily injection (MDI) therapy in patients requiring insulin treatment (4).

Taking the subject-specific glucose metabolism process as a controlled plant, the continuous glucose monitoring (CGM)-based MDI can be regarded as a multirate control generalization of the AP system with sparse insulin infusion, in the sense that the sampling rate of glucose measurements is much faster than that of insulin injection (ie, meal bolus and basal insulin). The underlying mathematical/theoretical basis of this approach is supported by the renowned multirate sampled-data control theory developed in control engineering (5, 6). With such a viewpoint, intelligent algorithm design for

MDI can benefit from the power of feedback intelligence, enabled by the legacy and advancement of systems and control theory. Although the titration of basal insulin in MDI therapy has been well studied, relatively little research has been carried out on preprandial insulin adjustment, which still relies on personal estimation of carbohydrate intake, insulin to carbohydrate ratio, and correction factors. Postprandial glucose control is also essential for achieving glycated hemoglobin A_{1c} (HbA_{1c}) goals (7), and the glucose level within 2 to 4 hours in the daytime is determined by just 1 shot of meal bolus in MDI therapy. Therefore, the design of intelligent learning and control algorithms for meal bolus decision in MDI is more technically challenging since it has stronger safety and robustness requirements. To mitigate this challenge, we proposed a data-driven risk-sensitive model-predictive control algorithm that seamlessly integrates Gaussian process glucose prediction models, Bayesian optimization, and event-triggered learning (8). This artificial-pancreas-like algorithm (AP-A) could recommend meal bolus insulin dosage through intelligent feedback control with reduced requirements on physiological parameters and clinical information (insulin to carbohydrate ratio and correction factors). The algorithm was able to make real-time preprandial insulin dosage recommendations based on glucose data from intermittently scanned continuous glucose monitoring (isCGM) and insulin injection information in the previous 24 hours (8). The algorithm operated in a closed-loop form, in the sense that it could respond to historically incorrect/inappropriate insulin doses and real-time feedback information offered by isCGM data based on the constantly updated prediction model, which is the major difference from existing meal bolus decision support systems (8).

Our preliminary study provided initial indications of the safety and clinical validity of using this algorithm in MDI therapy (8). The proposed approach was validated using both the Food and Drug Administration-accepted UVA/Padova simulator and compared with the standard insulin bolus calculator. Meanwhile, the high consistency rate between dosages recommended by the algorithm and those by the physicians in advisory-mode analysis using retrospective clinical data indicated the effectiveness and safety of the method (9). We hypothesized that adopting this AP-like algorithm in real-time preprandial insulin dosage decision guidance would result in noninferior glycemic control compared with dose adjustments performed by physicians from a specialized academic diabetes center.

Materials and Methods

Study Design and Participants

This single-blind, parallel (2-arm), randomized controlled, prospective, noninferiority trial was conducted at Peking University People's Hospital, Beijing, China, to evaluate the safety and efficiency of closed-loop meal bolus insulin dosage decision-making in MDI therapy on the percentage of time in the target range (TIR) (3.9–10 mmol/L) made using an AP-like algorithm (AP-A arm) compared with adjustments made by physicians from academic diabetes centers including certified endocrinologists. The trial was conducted in accordance with the International Declaration of Helsinki 2013 and International Conference on Harmonization Good Clinical Practice guidelines. The study was approved by the institutional review board at Peking University People's Hospital (2020PHB338-01). This study was registered May 8, 2021,

and January 7, 2022, with the China Medical Research Registration Information System (<https://www.nccl.org.cn>) and [Chictr.org](https://www.chictr.org.cn) (<https://www.chictr.org.cn>), under registration number MR-11-21-013314 and ChiCTR2200055328, respectively.

Eligible participants were aged at least 18 years, had an HbA_{1c} of at least 8.0%, and were in need of MDI therapy on admission (already using insulin or on highest dosage of sulfonylureas). Exclusion criteria were type 1 diabetes; any significant diseases including psychiatric, cardiovascular, liver, or kidney diseases; rejection or inability of using isCGM; unresolved skin conditions around the area of sensor placement; women of childbearing potential who were either pregnant at screening, or planning to become pregnant during the study period; use of medications that affect blood glucose within 1 month prior to enrollment (such as glucocorticoids). Participants were recruited from the pool of eligible people with type 2 diabetes who were hospitalized due to unsatisfactory glycemic control in the study center by the researching physician. All participants provided written informed consent.

Randomization and Masking

Eligible patients were randomly assigned (1:1) to the AP-A or physician arm. The randomization was stratified by age and HbA_{1c} at screening using a computer-generated random sequence, and had random block size. Participants were masked to the arm assignment, whereas physicians were not masked due to the nature of the intervention.

Procedures

Participants wore isCGM (Freestyle Libre H, Abbott) after random assignment to the AP-A or physician arm. In the AP-A arm, isCGM data were collected before meals and uploaded to the proposed closed-loop meal-bolus decision algorithm, which could automatically output a suggested insulin dosage. The physician would review the suggested dosage and choose either to adopt it or override the recommended dosage. Only 3 meal-bolus dosages (breakfast, lunch, and dinner) together with isCGM data 2 hours before and after each meal were needed for the algorithm to start providing dosage recommendations. In practice, isCGM data 2 hours before the meal should be uploaded to the algorithm before each recommendation. The actual administered dosage would also be sent back to the algorithm to update the data-driven insulin-glucose prediction model embedded in the control algorithm (7). In the physician arm, the insulin dosage adjustment was determined by certified endocrinologists according to current clinical strategies based on isCGM readings. For both arms, basal insulin dosage was also adjusted by the physicians in line with standard clinical practices (supplementary basal insulin dosage adjustment rules) (10). In addition, the starting insulin dosage was determined by the physicians based on the subject-specific history total insulin dosage, body weight, and glucose level at the beginning of the hospitalization (Supplementary Table S1) (10). Insulin lyspro (Humalog, Lilly) was used as the meal-time insulin and glargine U100 (Lantus, Sanofi) was used as the basal insulin in the study.

Clinical information was collected at baseline including age, sex, diabetes duration, comorbidities, complications, and concomitant medications. Height, weight, blood pressure, and waist circumference were measured the first day of hospitalization. After an overnight fast of 10 to 12 hours, fasted

venous blood (6 AM) was collected to measure fasting plasma glucose, creatinine, alanine transaminase, aspartate transaminase, uric acid, total cholesterol, triglycerides, high-density lipoprotein cholesterol (HDL-c), and low-density lipoprotein cholesterol (LDL-c) using a Hitachi LST008 analyzer (Hitachi High-Technologies). Fasting serum insulin and C-peptide levels were measured by chemiluminescence (Roche E411, Roche Diagnostics) using the Elecsys Insulin kit (catalog No. 07027559190, RRID: AB_2909455) and Elecsys C-Peptide kit (catalog No. 03184897190, RRID: AB_2909476). HbA_{1c} was assessed using high-performance liquid chromatography (Ultra2 HbA_{1c} Detector, Primus Corp). Urinary albumin/creatinine ratio was measured by Roche C311 (Roche Diagnostics).

A diagnosis of diabetic retinopathy was made according to the findings of direct ophthalmoscopy, fundus camera (TRC-NW100; Topcon Medical Systems) examinations, or a history of laser photocoagulation therapy. Coronary heart disease was diagnosed according to the patient's medical history (presence of angina or myocardial infarctions and the results of computed tomography imaging or coronary angiography). Cerebrovascular disease was defined as a history of ischemic or hemorrhagic stroke. Hypertension was defined as a systolic blood pressure of 140 mm Hg or greater and/or a diastolic blood pressure of 90 mm Hg or greater or current treatment of hypertension. Peripheral vascular atherosclerosis was determined via ultrasonography; if plaques were detected in the carotid artery or the arteries of the lower extremities, a diagnosis of peripheral vascular atherosclerosis was made.

The whole study was undertaken during hospitalization, and other antidiabetic medicines were continued after randomization except for sulfonylureas. A total of 6 endocrinologists were in charge of insulin dosage titration, and they were randomly assigned to the 2 arms according to the prearranged work schedule. Participants were provided with 3 meals per day, and the total calories were calculated based on age, sex, and body mass index (50% of the calories as carbohydrates, 30% lipids, and 20% proteins). The proportion of total daily calories for breakfast, lunch, and dinner were 20%, 40%, and 40%. However, the participants were allowed to take the amount of food by their own will and were also given the freedom to consume their own food during meals. Snacks with less than 15 g of carbohydrate (eg, 300 g of cucumber or tomato, <4 slices of soda biscuits) were also allowed during the study period.

Outcomes

The primary end point was the noninferiority for percentage time spent with sensor glucose level at 3.9 to 10.0 mmol/L (70–180 mg/dL) between the 2 study arms. The primary safety end point was the percentage time spent with sensor glucose level below 3.0 mmol/L (54 mg/dL). Secondary end points included percentage time spent with sensor glucose below 2.8 mmol/L (50 mg/dL), below 3.9 mmol/L (70 mg/dL), above 10 mmol/L (180 mg/dL), and above 13.3 mmol/L (240 mg/dL). Other end points were mean glucose, glucose variability evaluated by SD, coefficient of variation, glucose area under 10.0 mmol/L, and 3.9 mmol/L. Additionally, insulin dosage recommended by AP-A was also compared with the actual executed dosage, including agreement on insulin dosage, percentage time spent with sensor glucose at 3.9 to

10.0 mmol/L within 4 hours after physician overrides, and sensor glucose at 4 hours after physician overrides.

Data-driven Insulin Decision Algorithm

The intelligent insulin decision algorithm (namely, AP-A) was developed using a data-driven optimal control framework, which was motivated by a prediction-based optimization mechanism adopted using the model predictive control in artificial pancreas systems. Specifically, the algorithm was able to learn glucose dynamics and patterns by building a data-driven postprandial glucose prediction model in response to historical preprandial glucose traces, insulin dosages, and optional carbohydrate intakes based on a Gaussian-process, machine-learning method. With the provided multistep predictions (2 hours) of the postprandial glucose traces and the estimated uncertainties, an asymmetric stochastic cost function is constructed considering the asymmetric risks of hyperglycemia and hypoglycemia. A penalty term described by the square of the difference between the insulin dosage to be determined and the previous insulin dosage at the same meal time (breakfast, lunch, or dinner) is also incorporated. This enables the algorithm to quickly determine an appropriate dosage by finding a solution that minimizes the cost function in the neighborhood of the previous insulin dosage and is calculated by a Monte-Carlo-based Bayesian optimization approach. Meanwhile, the algorithm does not need carbohydrate counting and allows relatively flexible diet intake.

The algorithm was designed to provide insulin dosage recommendations without the estimations of insulin to carbohydrate ratio and correction factors, and requires only 3 meal-related data samples (breakfast, lunch, and dinner) to produce recommendations. A valid sample should include the preprandial isCGM data (2 hours), the corresponding insulin dosage, and postprandial isCGM data (2 hours). With the help of the robust modeling capability of the Gaussian-process machine learning, the algorithm is robust to carbohydrate-counting disturbances within a tolerable range. The isCGM data (at least 2 hours) are collected 15 minutes before the meal and fed to the algorithm to produce recommendation for each meal. The recommended dosage was rounded and then reviewed (approved or revised) by the physicians. When insulin administration was completed, the new produced sample would be included in the training sets of the prediction model. For safety concerns, the postprandial glucose target at 2 hours after a meal is tailored for each participant according to his or her age and diabetes comorbidities/complications (Supplementary Table S2) (10). Moreover, the dosage change is limited to ± 3 U of the previous insulin dosage at the same meal time, and the physician-logic based if-then rules are incorporated to perform decisions in case the prediction model cannot effectively capture the effect of insulin on postprandial glucose (eg, lowering the glucose level by increasing insulin dosage) to enhance safety.

Statistical Analysis

We hypothesized that the AP-A-guided arm would not be inferior to the physician arm in percentage of glucose readings within the target range of 70 to 180 mg/dL (3.9–10 mmol/L). The sample size was calculated using PASS15.0 for noninferiority analysis, with a margin of 7.5% between arms in a parallel study design with power of 90%, a one-sided significance level of 5%, and SD of 13% according to previous

reports (11, 12). A sample size of 106 participants was required for the study, and considering a 5% dropout rate, the sample size was finally set at 111 participants.

The analyses were performed in the intention-to-treat participants. The per-protocol analysis was consistent with the intention-to-treat analysis since the drop-off ($N=2$) in the control group was due to sensor loss. The coprimary end points used a stepdown strategy designed to maintain type I error at 5% or less for multiple comparisons. The AP-A-guided arm was considered noninferior compared to the control arm if coprimary end points were met. A noninferiority test was conducted for the other end points using a randomization test with a significance level of 5%. The noninferiority was assessed based on the defined margin. For the end points related to hypoglycemia and describable by TIRE, the noninferiority limits were all selected as 2% (8). The secondary end points were then tested in a prespecified order until one of the null hypotheses was nonrejected. The physicians' overrides analysis was computed as the percentage of change from the original algorithm recommendations. The statistical analyses for baseline characteristics of participants were conducted using SPSS, version 27 (IBM Software). Other statistical analyses were performed using MATLAB 2017b (Mathworks Inc) for Windows and R, version 4.2.1 (R Foundation for Statistical Computing). Statistical significance was defined as P values less than .05 with 2-sided testing.

Results

From November 6, 2021 to July 26, 2022, 140 patients were assessed for eligibility; 119 were enrolled and randomly assigned to the AP-A arm ($n=59$) or the physician arm ($n=60$). All participants in the AP-A arm and 58 of 60 participants in the physician arm completed the study (Fig. 1).

The demographic and clinical characteristics at baseline were well balanced between the 2 study arms (Table 1). The mean age in the AP-A arm was 60.8 ± 13.15 years and 60.3 ± 12.1 years in the control arm ($P=.842$). Diabetes duration was 14.5 ± 13.0 years in the AP-A arm and 13.7 ± 10.3 years in the control arm ($P=.621$). HbA_{1c} was $10.1 \pm 1.9\%$ in the AP-A arm and $10.1 \pm 2.1\%$ in the control arm ($P=.948$). No statistical differences were observed for other baseline characteristics between the 2 arms (see Table 1).

The median (minimum-maximum) run time of the algorithm in the AP-A arm was 3.25 (1.25-8.25) days, and the parallel insulin decision times were 10 (4-25) times. On the first day of the run time, the algorithm was trained only by the isCGM data and insulin dosage without providing insulin dosage recommendations. The analysis of the outcomes was performed using the data collected during the whole MDI therapy period, including the first 24 hours. The primary efficacy end point, TIR (3.9-10.0 mmol/L), achieved by the AP-A arm, was statistically noninferior compared with the control arm (72.4% [63.3%-82.1%] vs 71.2% [54.9%-81.4%]), with a median difference of 1.33% (95% CI, -6.00 to 10.94). The primary safety end point, TBR (<3.0 mmol/L), was also statistically noninferior between the AP-A and control arms (0.0% [0.0%-0.0%] vs 0.0% [0.0%-0.0%]), respectively; median difference (95% CI) 0.00% (0.00%-0.00%). All other secondary end points in the AP-A arm were noninferior compared with the physician arm (Figs. 2 and 3). No episodes of hyperglycemia with diabetic ketoacidosis or severe hypoglycemia occurred in either group during the study period.

The median TIR in AP-A arm was 58.8% ($N=59$) the first day and progressively increased to 92.7% ($N=7$) on the sixth day, whereas the median TIR in the physician arm was 60.4% ($N=58$) the first day and progressively increased to 78.1% ($N=12$) on the sixth day (Supplementary Fig. S1) (10). The median (first quartile-third quartile) preprandial insulin dosage per day throughout the titration period in the AP-A arm was 22 (18-28), 21 (18-25), 21 (15-26), 15 (10-24), 17 (6-33), and 18 (13-32) units, and the median (first quartile-third quartile) basal insulin dosage per day in the AP-A arm was 14 (11-18), 16 (12-19), 14 (10-18), 14 (11-18), 15 (10-22), and 18 (15-20) units. The median (first quartile-third quartile) preprandial insulin dosage per day in the physician arm was 20 (15-24), 21 (17-26), 24 (18-30), 24 (18-31), 15 (8-26), and 26 (19-35) units, and the median (first quartile-third quartile) basal insulin dosage in the physician arm was 12 (10-16), 14 (10-16), 14 (10-18), 14 (10-17), 16 (13-22), and 20 (15-24) units during the study.

The total number of AP-A recommendations made for participants in AP-A arm were 412, of which 257 (62.4%) recommendations were consistent with actual administration. Among the 155 overrides, 79 were kept the same with the previous insulin dosages at the same meal time (breakfast, lunch, or dinner) before the algorithm recommendations (mainly due to technical issues that were not related to the algorithm, including failure of uploading isCGM data in time, unable to send recommended dosage in a timely manner, and unable to implement recommended dosage), 21 were overridden in the same direction as the algorithm recommendations but with different magnitudes, 11 were changed when the algorithm did not make a changed recommendation, 36 were unchanged while the algorithm made a change recommendation, and 8 were changed in the opposite direction of the algorithm recommendation. The plots of the consistency between proposed and actual administered insulin dosage are provided in Fig. 4, which shows that 92.2% of the differences were within the 95% limits of agreement (-2.66 to 2.56 U). Fig. 5 provides details about acceptance and physician overrides from day 1 to day 8, which suggested that even during the first day of recommendation, 52.0% of insulin dosage recommended by AP-A were exactly the same as the actual administered dosage.

To further explore the safety of the algorithm, we divided the 412 dosages suggested by AP-A into 3 parts: recommendations that were not overridden ($n=253$), recommendations that were overridden so that insulin was reduced by the physician relative to the AP-A recommendation ($n=76$), and recommendations that were overridden so that insulin was increased by the physician relative to the AP-A recommendation ($n=77$) (Table 2). For the recommendations that were overridden so that insulin was reduced by the physician relative to the AP-A recommendation, TIR within 4 hours after insulin administration was similar in both the circumstances: recommendations not overridden in the AP-A arm, and those in the physician arm (62.5% [18.8%-96.9%] vs 62.5% [28.1%-100%], 62.5% [18.8%-96.9%] vs 56.3% [12.5%-100%]). The sensor glucose at 4 hours after insulin administration was also the same in such comparisons (8.4 mmol/L [6.6-10.6] vs 8.3 mmol/L [6.7-10.6], 8.4 mmol/L [6.6-10.6] vs 8.5 mmol/L [6.6-11.3]). These comparisons suggested that the performance of the algorithm was similar to the physicians with a high safety guarantee. During the circumstances in which recommendations were overridden so that

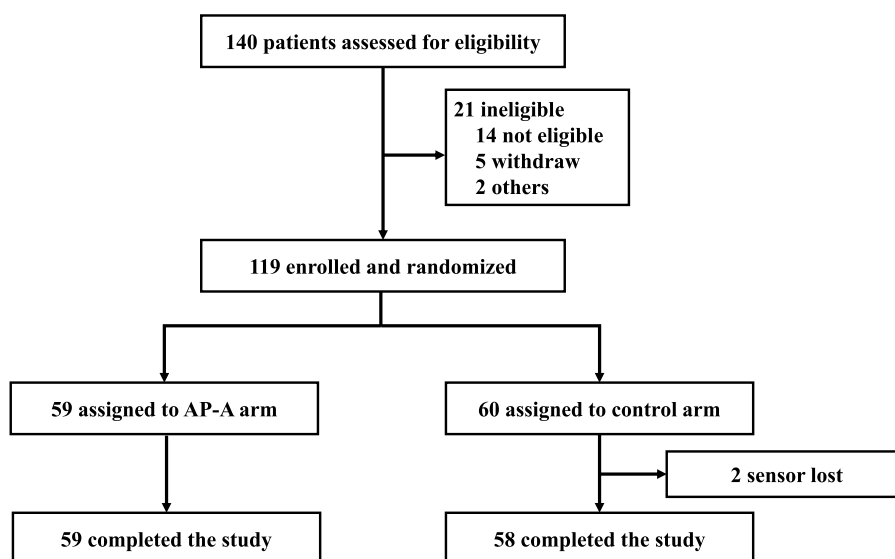


Figure 1. Trial profile. No participants in the artificial-pancreas-like algorithm (AP-A) arm discontinued and 2 participants in the physician arm discontinued during the study period.

insulin was increased by the physician relative to the AP-A recommendation, TIR within 4 hours after insulin administration was statistically better when comparing to both circumstances: recommendations not overridden in the AP-A arm (81.3% [56.3%-100%] vs 62.5% [28.1%-100%]; $P = .001$) and those in the physician arm (81.3% [56.3%-100%] vs 56.3% [12.5%-100%]; $P < .001$). Meanwhile, a lower sensor glucose level at 4 hours after physician overrides was also observed in such comparisons (7.0 mmol/L [5.3-9.1 mmol/L] vs 8.3 mmol/L [6.7-10.6 mmol/L], 7.0 mmol/L [5.3-9.1 mmol/L] vs 8.5 mmol/L [6.6-11.3 mmol/L]; both $P < .001$). These comparisons suggest that the performance of the algorithm was relatively conservative during cases of low sensor glucose level, while in the AP-A arm physician decisions on titration were more aggressive. Additionally, the analysis of TIR, TBR, and other secondary end points excluding those physician override periods also suggested noninferiority compared with the physician arm (Supplementary Table S3) (10).

Discussion

Our study is the first randomized controlled trial to evaluate the efficacy and safety of adopting an AP-like closed-loop intelligent-decision algorithm in preprandial insulin dosage titration during MDI therapy. We found that the use of the AP-like algorithm for dosage titration guidance was noninferior in efficacy and safety compared to care provided by specialists. The AP-like algorithm was found to be safe, with the percentage of readings in range (3.9-10.0 mmol/L [70-180 mg/dL]) and below range (3.0 mmol/L [54 mg/dL]) being statistically noninferior to that in the physician arm.

Frequent and adequate insulin titration is essential to ensure the effectiveness of insulin therapy (13, 14), but hard to achieve in clinical practice. The evolving technology of the artificial pancreas is aiding in closing the titration gap. APs provided a more physiologic approach to diabetes management by tying insulin delivery to sensor glucose values on the basis of closed-loop controlled algorithms. Insulin delivery with such devices can be interrupted to avert hypoglycemia, and additional insulin can be automatically delivered to

mitigate hyperglycemia. The benefits of automated insulin delivery have become clear, as evidenced by improvements in glycemic control, particularly during overnight hours (15). However, because of the cost and complexity of such devices, most insulin users with type 2 diabetes are unlikely to use APs. Automating MDI can definitely be a more cost-effective and widely adopted option, but more advanced and dedicated algorithms need to be developed, particularly considering the challenges in relatively longer-term glucose prediction than AP and higher risks in deciding insulin dosages, which normally have larger size than those of the AP microboluses. Through learning the subject-specific postprandial insulin-glucose dynamics and meal-intake patterns with a data-efficient Gaussian process prediction model, the AP-like algorithm is able to perform insulin to carbohydrate ratio and correction factor-free preprandial insulin dosage suggestions. The safety of the meal bolus dosage decision is ensured by designing a risk-sensitive cost function in the model predictive control optimization problem, together with clinical experience-enabled safety constraints. By updating the prediction model for every meal and performing glucose predictions based on the premeal isCGM trajectories, the algorithm is able to achieve closed-loop meal bolus recommendations like the feedback control algorithms used in an AP. Although glucose variability in patients with insulin-treated type 2 diabetes was lower compared with patients with type 1 diabetes (16), attempts were also made to see the benefits of fully closed-loop insulin delivery system in adults with type 2 diabetes (17, 18), which suggested that the personalization of insulin delivery improved glucose control comparing with standard insulin therapy. Nowadays, titration of basal insulin has been well studied and is relatively easy to conduct both for providers and patients. However, proper preprandial insulin dose adjustments rely on estimation of carbohydrate intake, insulin to carbohydrate ratio, and correction factors. Asia has the largest diabetes population worldwide and starchy foods are the main source of dietary energy (19), accompanied by the tradition of shared dishes in most parts of Asia, preprandial insulin titration is extremely difficult, especially for those who do not have the required expertise and

Table 1. Baseline characteristics

	AP-A arm (n = 59)	Physician arm (n = 60)	P
Sex, male	40 (68%)	31 (52%)	.073
Age, y	60.8 (13.2)	60.3 (12.1)	.842
Age at diagnosis, y	46.1 (13.0)	46.9 (12.9)	.742
Diabetes duration, y	14.5 (9.0)	13.7 (10.3)	.621
Height, m	1.7 (0.1)	1.6 (0.1)	.137
Weight, kg	73.9 (11.1)	70.1 (11.4)	.068
BMI	27.0 (3.4)	26.4 (3.5)	.334
Waist circumference, cm	96.9 (8.4)	94.4 (9.6)	.138
SBP, mm Hg	138 (17)	143 (22)	.146
DBP, mm Hg	79 (12)	79 (14)	.691
HbA _{1c} , %	10.1 (1.9)	10.1 (2.1)	.948
GA, %	25.8 (7.2)	25.9 (7.2)	.909
Fasting plasma glucose, mmol/L	9.0 (2.8)	9.3 (3.5)	.577
Glucose (120 min), mmol/L	12.4 (3.4)	13.0 (5.2)	.411
C-peptide (fasting), ng/mL	1.8 (1.4-2.4)	1.7 (1.3-2.8)	.966
C-peptide (120 min), ng/mL	3.0 (1.7-4.4)	3.1 (1.9-5.2)	.597
Fasting insulin, μ IU/mL	7.6 (4.7-10.8)	7.9 (5.1-14.2)	.311
Insulin (120 min), μ IU/mL	16.1 (9.0-32.8)	16.7 (10.5-35.4)	.355
TC, mmol/L	4.7 (2.4)	4.5 (1.5)	.719
TGs, mmol/L	2.3 (2.5)	1.8 (1.5)	.124
HDL, mmol/L	1.1 (0.3)	1.1 (0.3)	.129
LDL, mmol/L	2.9 (1.3)	2.8 (1.1)	.481
ALT, U/L	22.5 (14.9)	20.4 (11.4)	.378
AST, U/L	20.2 (8.6)	19.7 (7.8)	.697
UA, μ mol/L	364.3 (86.0)	340.0 (81.1)	.117
CRE, μ mol/L	75.0 (41.6)	66.8 (22.5)	.186
eGFR, mL/min per 1.73 m ²	91.3 (25.5)	93.7 (20.3)	.568
UACR, mg/g	13.6 (4.9-75.9)	11.1 (6.0-36.4)	.340
Comorbidity			
Hypertension	39 (66%)	41 (68%)	.795
Coronary atherosclerotic heart disease	10 (17%)	13 (22%)	.515
Cerebrovascular disease	1 (2%)	1 (2%)	.990
Diabetic retinopathy	22 (37%)	21 (35%)	.795
Peripheral vascular disease	56 (95%)	56 (93%)	.714
Medication			
Statins	51 (83%)	47 (80%)	.326
Blood pressure-lowering medication	39 (66%)	39 (66%)	1.000
Oral antidiabetic medications			
Metformin	37 (63%)	30 (52%)	.230
SGLT-2i	12 (20%)	9 (16%)	.497
DPP-IV inhibitors	11 (19%)	12 (21%)	.781
GLP-1	5 (8%)	4 (7%)	.749
Acarbose	17 (29%)	17 (29%)	.953
Sulfonylureas	12 (21%)	10 (17%)	.636
TZD	2 (3%)	1 (2%)	.569
Insulin			

(continued)

Table 1. Continued

	AP-A arm (n = 59)	Physician arm (n = 60)	P
Basal insulin	14 (24%)	13 (22%)	.866
Premix insulin	10 (17%)	10 (17%)	.967
MDI	17 (29%)	10 (17%)	.137

Data are mean (SD), median (first quartile-third quartile) or N (%) unless otherwise indicated.

Three individuals were missing fasting insulin; 3 individuals were missing insulin (120 minutes); 1 participant was missing C-peptide (120 minutes); 1 individual was missing eGFR; 3 individuals were missing information about ultrasonography results; and 1 individual was missing medication used for comorbidities. Multiple imputation was used to supplement missing values. Statistical hypothesis testing *P* greater than .05 (2-tailed) was considered as no statistically significant difference. Abbreviations: ALT, alanine transaminase; AP-A, artificial-pancreas-like algorithm; AST, aspartate transaminase; BMI, body mass index; CRE, creatinine; DBP, diastolic blood pressure; DPP-IV, dipeptidyl peptidase IV; eGFR, estimated glomerular filtration rate; GA, glycated albumin; GLP-1, glucagon-like peptide-1; HbA_{1c}, glycated hemoglobin A_{1c}; HDL, high-density lipoprotein; LDL, low-density lipoprotein; MDI, multiple daily injections; SBP, systolic blood pressure; SGLT-2i, sodium-glucose cotransporter-2 inhibitors; TC, total cholesterol; TGs, triglycerides; TZD, thiazolidinedione; UA, uric acid; UACR, urinary albumin to creatinine ratio.

skills in calculating premeal insulin dosage on a per-meal basis. Using an AP-like algorithm in supporting meal-time insulin dosage decision would help to overcome the 2 main obstacles in appropriate titration, clinical inertia and fear of hypoglycemia (20).

A key strength of our study was the use of a noninferiority design to evaluate the new AP-like algorithm by comparison with a standard clinical approach provided by diabetes specialists from academic centers. Another strength of our study was that the population recruited in the study had relatively long diabetes duration and a high prevalence of cardiovascular comorbidities, and our proposed algorithm was able to achieve both the safety and efficacy end points. Furthermore, the high compliance rate and completion rate also suggested the feasibility of adopting this algorithm for guiding insulin titration in the clinical setting.

Several alternative approaches to optimizing insulin therapy exist. As digital communication technology advances, phone calls, video calls, and smartphone applications have made remote control of glucose possible (21), and the use of these technologies may be even more important in situations such as the COVID-19 pandemic, during which access to care providers becomes more challenging. Nowadays, many titration guidelines have been transformed into smartphone applications that are used by either patients or providers, and telemedicine platforms could allow providers to have phone call or video calls with patients and to make medication adjustments (22). However, such solutions still require health care providers, and since time is already overstretched for them, it is unrealistic for them to perform real-time insulin dosage titration for patients. Investigations about these platforms have suggested conflicting results, and meta-analysis shows that these may be considered as an adjunct to improve glycemic control only when acceptability and functionality have been taken into consideration (23). Meanwhile, despite the long-term availability of these insulin titration platforms, glycemic control in patients who are treated with insulin has not improved for decades (24). Another method for insulin titration is by run-to-run algorithms based on self-monitoring of blood glucose results, and now such devices

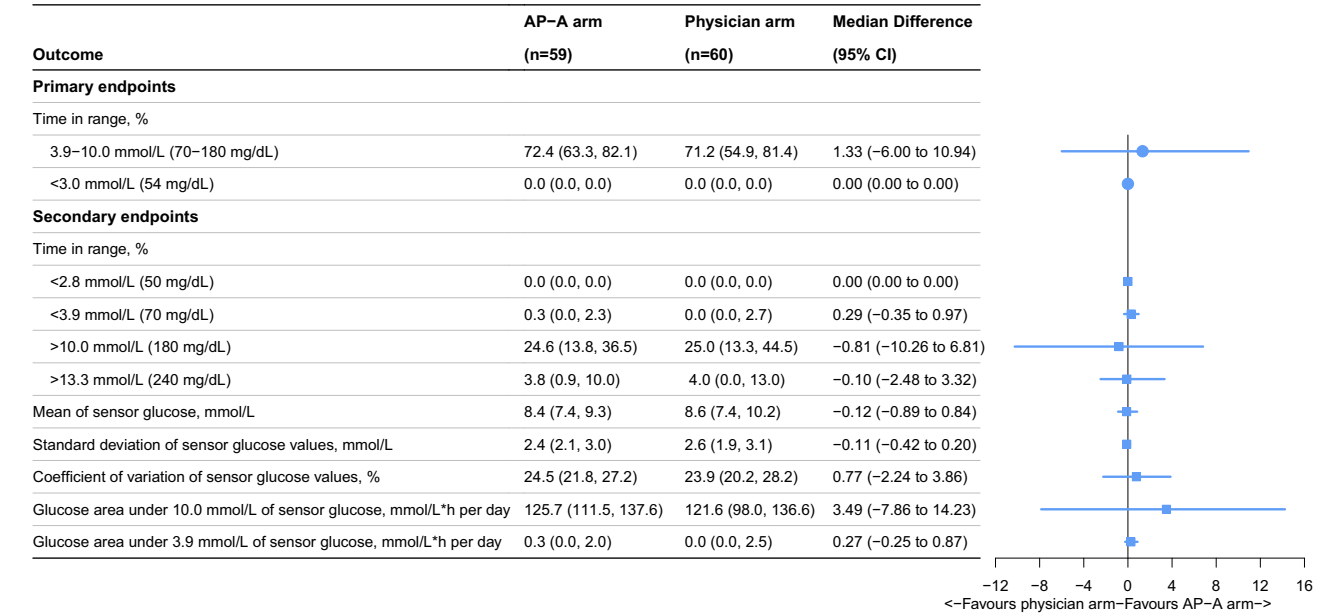


Figure 2. Intermittently scanned continuous glucose monitoring (isCGM)-derived end points of glycemic control. Data are median (first quartile, third quartile) unless otherwise indicated. Median difference was calculated from the differences between the artificial-pancreas-like algorithm (AP-A) arm and the physician arm. Noninferiority is met when the upper confidence limit is less than the noninferiority margin or when the lower confidence limit is greater than the noninferiority margin.

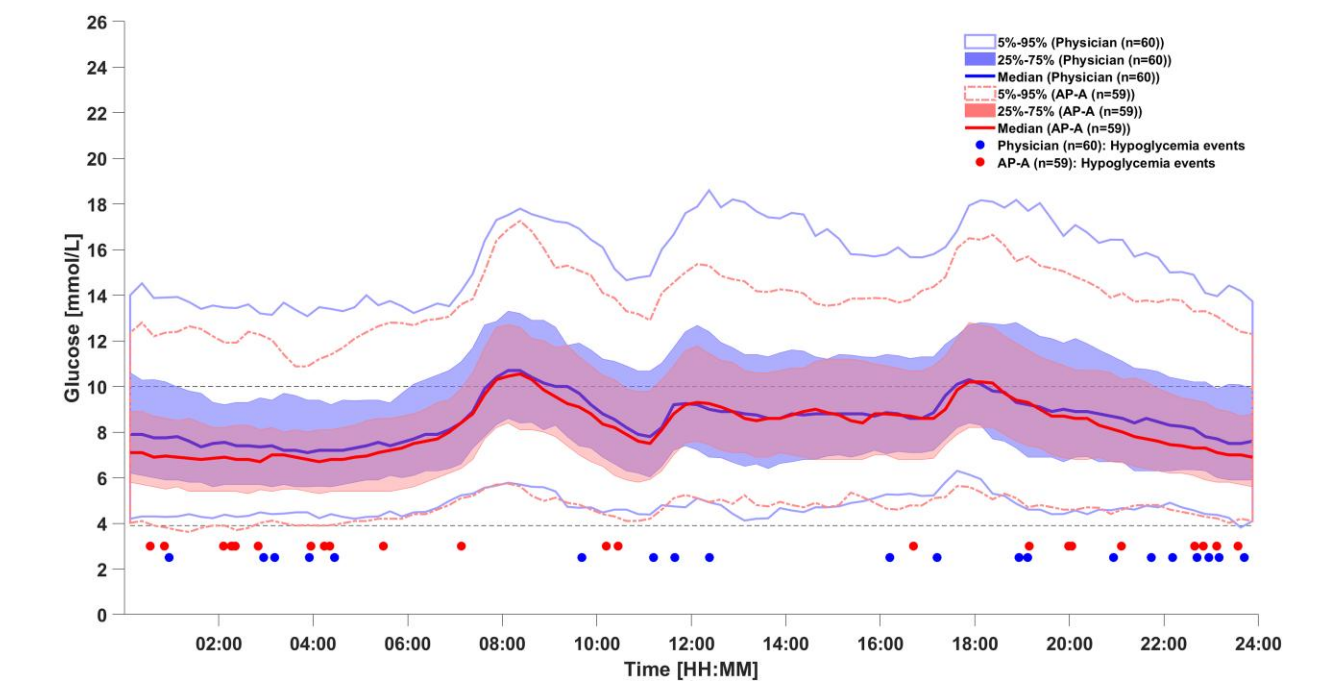


Figure 3. Intermittently scanned continuous glucose monitoring (isCGM) glucose pattern during the study period. Line and solid area in the lower position, and area enclosed by dashed line denote the median, 25% to 75%, and 5% to 90% of sensor glucose level in the artificial-pancreas-like algorithm (AP-A) arm. Line and solid area in the up position, and area enclosed by solid line denoted the median, 25% to 75%, and 5% to 90% of sensor glucose level in the physician arm. Dots in the top row denoted the hypoglycemia events (<3.0 mmol/L [54 mg/dL]) in the AP-A arm. Dots in the lower row denoted the hypoglycemia events (<3.0 mmol/L [54 mg/dL]) in the physician arm.

are commercially available in some countries. Although short-term clinical trials proved the efficacy and safety of such devices (25), long-term and multiregion trials are still lacking. Many data delivery systems enable real-time delivery of CGM glucose readings from their monitoring devices to their providers (26). Although some attempts have also been made by using CGM data to guide insulin titration,

nearly all of them were performed in those using continuous insulin infusion treatment (27–29), and none of them used the closed-loop control algorithms in guiding titration in MDI therapy. To our best knowledge, our study was the first randomized controlled trial study to prove the feasibility, efficacy, and safety of using an AP-like algorithm in preprandial insulin dosage titration.

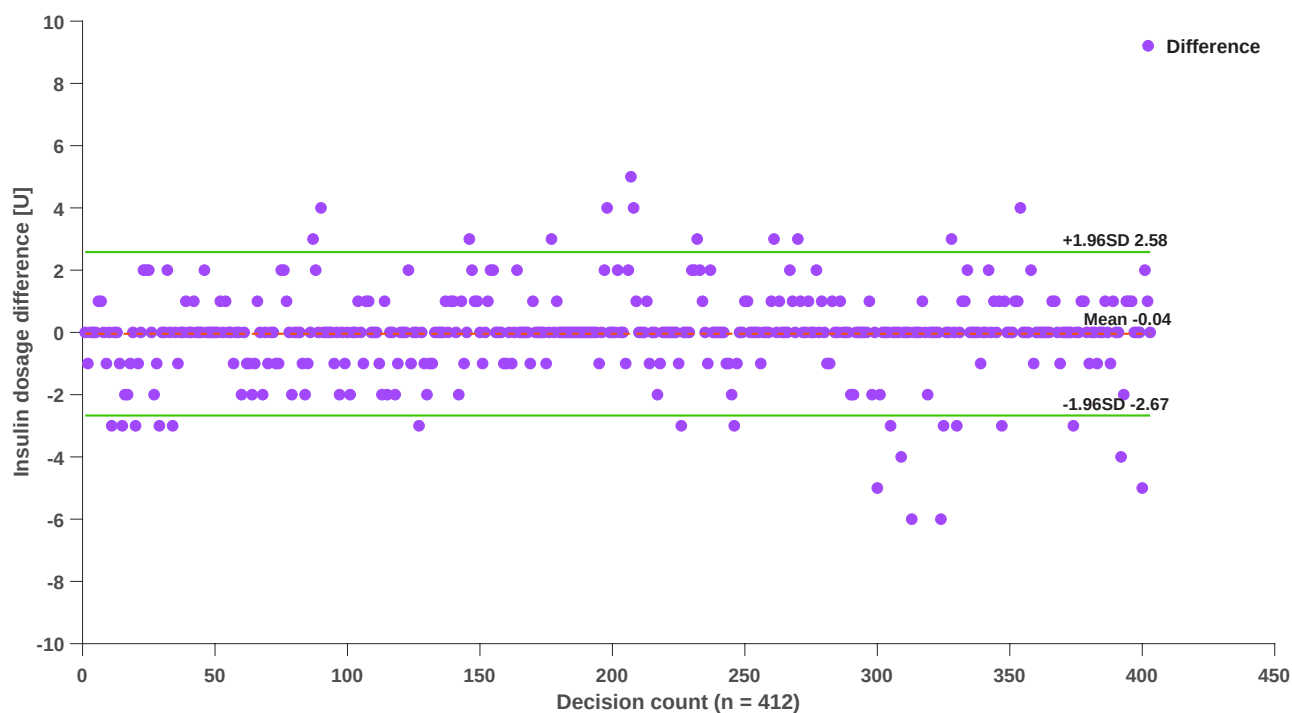


Figure 4. Bland-Altman plots of the agreement between proposed and actual administered insulin dosage. The differences between proposed and actual administered insulin dosage are represented as dots. The mean difference between proposed and actual administered insulin dosage is presented by the dotted line (-0.05 U). The 95% limits of differences (-2.66 to 2.56 U) are denoted by the zone enclosed by the 2 solid lines. A total of 92.2% of the differences fell in the 95% limits.

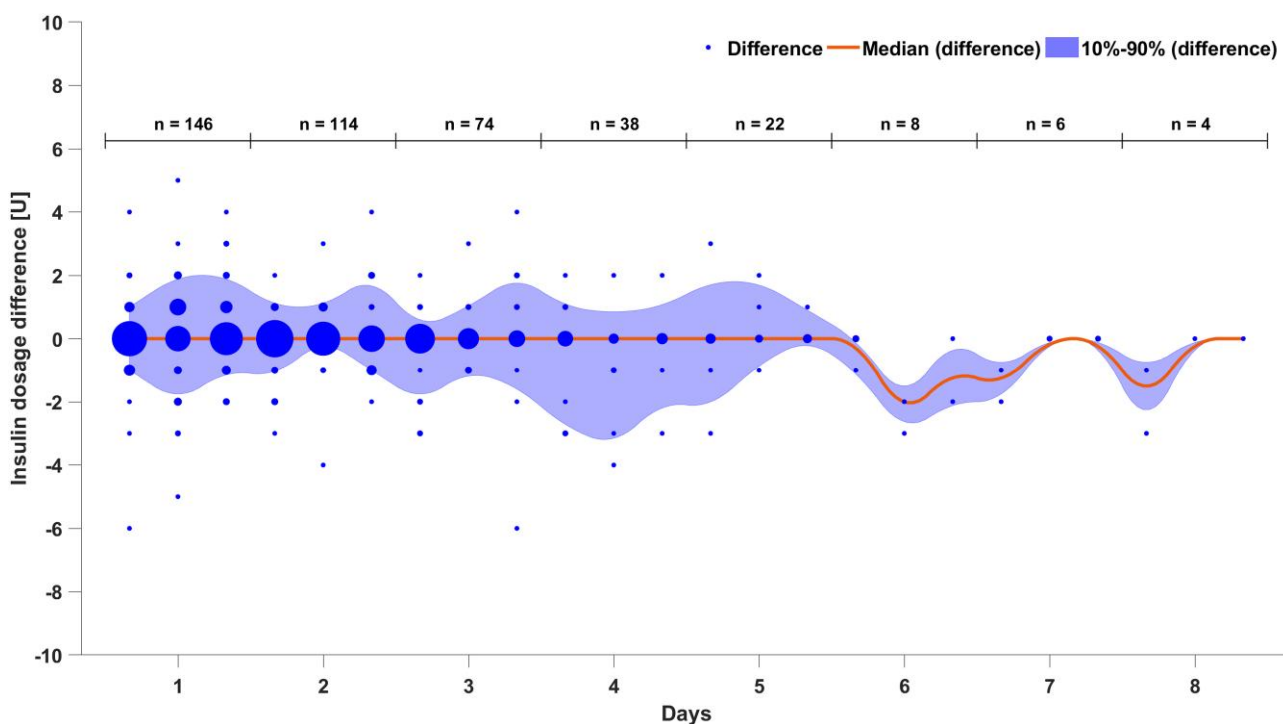


Figure 5. Difference between proposed and actual administered insulin dosage across the decision days. The differences between proposed and actual administered insulin dosage are represented as dots. The sizes of the dots were proportional to their frequency. The line (smoothed) denotes the median value curve. The solid area denotes the zone enclosed by the 10% and 90% quartile line (smoothed) of the difference. Most (81%) of the differences between the proposed and actual administered insulin dosage on the first day fell between -1 and 1 U.

A potential limitation is that the override rate was relatively nonnegligible. However, the fact that some physicians overrode the AP-A recommendations is not surprising. Insulin

dose adjustments are not only influenced by science but also by different subjective attitudes, and individual expert clinicians have been shown previously to provide different insulin

Table 2. Evaluation results of insulin decision differences

Insulin decision results	AP-A arm (N = 59)			Physician arm (N = 60)
Average agreement rate, %	65.2 (19.3)			N/A
Agreement on all recommendations, %	62.4 (257/412)			N/A
isCGM-derived evaluation results	Case of recommended insulin dosage consistent with actual administered one (n = 253)	Case of recommended insulin dosage > actual administered one (n = 76)	Case of recommended insulin dosage < actual administered one (n = 77)	Case of insulin dosage determined by physicians (n = 637)
Time in range of 3.9–10.0 mmol/L within 4 h after insulin bolus, %	62.5 (28.1–100)	62.5 (18.8–96.9)	81.3 (56.3–100) ^{a,b}	56.3 (12.5–100)
Sensor glucose at 4 h after insulin bolus, mmol/L	8.3 (6.7–10.6)	8.4 (6.6–10.6)	7.0 (5.3–9.1) ^{a,b}	8.5 (6.6–11.3)

Data are median (first quartile–third quartile), mean (SD) or % (n/N), unless otherwise indicated.

Average agreement rate was defined as average dose acceptance rate determined by each participant. Agreement on all recommendations was defined as acceptance rate of overall recommended dosages.

Abbreviations: AP-A, artificial-pancreas-like algorithm; isCGM, intermittently scanned continuous glucose monitoring.

Results with

^a differed significantly from the results in the case of recommended insulin dosage consistent with the actual administered one. Results with

^b differed significantly from the results in the physician arm. Statistical significance was assessed by Mann-Whitney *U* test. *P* less than .05 (2 tailed) was considered statistically significant.

dosing adjustments to the same patient given patient data (30). Meanwhile, most of the physician overrides in our study were due to technical problems unrelated to the algorithm, such as being unable to upload isCGM data in time, or a missed transmission of insulin dosage. And the analysis excluding those override periods suggested the same results. The analysis about the glucose level after overrides also suggested that AP-A recommendations were conservative and restrictive, indicating that the physician overrides were aiming for better glucose control, and under the guidance of the algorithm, endocrinologists were able to facilitate a better glucose target. Since many investigations today show that TIR is associated with diabetes complications (31, 32), it is valid to consider that adopting this AP-A supporting insulin titration would benefit patients under MDI therapy.

Another limitation of our study was its relatively short duration and its single-center design. Since diabetes is a chronic disease and most of the titration would be performed by patients alone in an at-home setting, studies that evaluate long-term, multicenter, home use of the AP-A during insulin titration will be needed. The main strengths of the study are the large number of participants, and that it first proved the concept of adopting an AP-like algorithm in MDI insulin titration. Meanwhile, since the insulin titration was performed by experienced endocrinologists in the control arm, noninferiority achieved by AP-A guidance is promising for its further use.

The third limitation pertains to the necessity for precise insulin dosage titration in type 1 diabetes. In our proposed study, we are considering integrating CGM with Bluetooth technology and a smart insulin pen. This integration, guided by AP-A, will establish a wireless AP-A insulin dosage-adjustment system for glucose control in type 1 diabetes.

In summary, AP-A guidance of preprandial insulin titration was noninferior in both efficacy and safety compared to physician care alone in type 2 diabetes patients treated with an MDI regimen.

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Author Contributions

W.L., .DC., D.S., and L.J. designed the trial and performed data analysis. R.Z., X.Z., X.C., M.L., W.W., and Y.L. contributed to data collection. L.T. supervised the statistical process. All authors interpreted the data and were involved in the conducting of the trial. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication. All authors contributed to the data interpretation and manuscript writing, approved the final version of the manuscript, and vouch for the data accuracy and fidelity to the protocol.

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Disclosures

The authors have nothing to disclose.

Data Availability

All data sets generated during and analyzed during the current study are not publicly available but are available from the corresponding author on reasonable request. Data will be shared with bona fide researchers who submit a research proposal approved by the independent review board. Individual

participant data will be shared in data sets in a deidentified and anonymized format.

Clinical Trial Information

This study was registered both with the China Medical Research Registration Information System and Chictr.org, number MR-11-21-013314 and ChiCTR2200055328, respectively (registered May 8, 2021 and January 7, 2022, respectively).

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